

Exercise 4 of module 1004D on "Theoretical spectroscopy for systems implying periodic boundary conditions"**Second quantization****Exercise 1**

Creation and annihilation operators, $\hat{a}_i^\dagger$ and $\hat{a}_i$, acting onto the Slater determinant $|\chi_k \dots \chi_l\rangle$ are defined as

$$\hat{a}_i^\dagger |\chi_k \dots \chi_l\rangle = |\chi_i \chi_k \dots \chi_l\rangle, \quad (1)$$

$$\hat{a}_i |\chi_i \chi_k \dots \chi_l\rangle = |\chi_k \dots \chi_l\rangle. \quad (2)$$

Apply the following operator expressions onto a Slater determinant:

$$\text{a) } \hat{a}_i^\dagger \hat{a}_i^\dagger \quad (3)$$

$$\text{b) } [\hat{a}_i^\dagger, \hat{a}_j^\dagger]_+ = \hat{a}_i^\dagger \hat{a}_j^\dagger + \hat{a}_j^\dagger \hat{a}_i^\dagger \quad (4)$$

$$\text{c) } [\hat{a}_i^\dagger, \hat{a}_j]_+ = \hat{a}_i^\dagger \hat{a}_j + \hat{a}_j \hat{a}_i^\dagger \quad (5)$$

$$\text{d) } \sum_i \hat{a}_i^\dagger \hat{a}_i \quad (6)$$

Exercise 2

The one-electron operator $\hat{f}$ and two-electron operator $\hat{g}$ are defined as

$$\hat{f} = \sum_{p\sigma_p q\sigma_q} f_{p\sigma_p q\sigma_q} \hat{a}_{p\sigma_p}^\dagger \hat{a}_{q\sigma_q}, \quad (7)$$

$$\hat{g} = \frac{1}{2} \sum_{pqrs} \sum_{\sigma_p \sigma_q \sigma_r \sigma_s} g_{p\sigma_p q\sigma_q r\sigma_r s\sigma_s} \hat{a}_{p\sigma_p}^\dagger \hat{a}_{r\sigma_r}^\dagger \hat{a}_{s\sigma_s} \hat{a}_{q\sigma_q}, \quad (8)$$

with $p, q, r, s \dots$ summing over all spin orbitals with corresponding spin σ . Show that the spin-free one-electron and two-electron operators are given as

$$\hat{f} = \sum_{pq} f_{pq} \hat{E}_{pq} \text{ with } \hat{E}_{pq} = \sum_{\sigma} \hat{a}_{p\sigma}^\dagger \hat{a}_{q\sigma}, \quad (9)$$

$$\hat{g} = \frac{1}{2} \sum_{pqrs} g_{pqrs} \hat{e}_{pqrs} \text{ with } \hat{e}_{pqrs} = \hat{E}_{pq} \hat{E}_{rs} - \delta_{qr} \hat{E}_{ps}. \quad (10)$$

Exercise 3

Verify that

$$[\hat{H}, \hat{E}_{ai}] = \sum_p h_{pa} \hat{E}_{pi} - \sum_p h_{ip} \hat{E}_{ap} + \sum_{pqr} g_{pqra} \hat{e}_{pqri} - \sum_{pqr} g_{pqir} \hat{e}_{pqar}, \quad (11)$$

and that

$$\begin{aligned} [\hat{H}, \hat{E}_{ai}]|\text{HF}\rangle &= 2F_{ia}|\text{HF}\rangle + \left(\sum_b F_{ba} \hat{E}_{bi} - \sum_j F_{ij} \hat{E}_{aj} + \sum_{bj} L_{bjia} \hat{E}_{bj} \right) |\text{HF}\rangle \\ &+ \left(\sum_{bjc} g_{bjca} \hat{E}_{bj} \hat{E}_{ci} - \sum_{bjk} g_{bjik} \hat{E}_{bj} \hat{E}_{ak} \right) \end{aligned} \quad (12)$$

with

$$\hat{H} = \sum_{pq} h_{pq} \hat{E}_{pq} + \frac{1}{2} \sum_{pqrs} g_{pqrs} \hat{e}_{pqrs} + V_{\text{nuc}}, \quad (13)$$

$$F_{pq} = h_{pq} + \sum_i (2g_{pqii} - g_{p i i q}), \quad (14)$$

$$L_{pqrs} = (2g_{pqrs} - g_{psrq}). \quad (15)$$

Exercise 4

Verify that the overlap of two doubly excited states is given as

$$\langle_{ij}^{ab} |_{kl}^{cd} \rangle = 4\delta_{ac}\delta_{bd}\delta_{ik}\delta_{jl} + 4\delta_{ad}\delta_{bc}\delta_{il}\delta_{jk} - 2\delta_{ac}\delta_{bd}\delta_{il}\delta_{jk} - 2\delta_{ad}\delta_{bc}\delta_{ik}\delta_{jl}. \quad (16)$$